

Calculator Project

Concept: I wanted to create a project that combined my interest in arithmetic with an opportunity to practice my web design skills. Inspired by beginner-friendly projects, I turned to a YouTube tutorial titled "How To Make A Calculator Using HTML CSS And JavaScript" by GreatStack. Using knowledge gained from personal HTML, CSS, and JavaScript courses, I followed along with the tutorial and customized the base calculator design. Once I completed the tutorial, I felt motivated to improve and expand upon the project by adding new features and enhancements to make it uniquely mine.

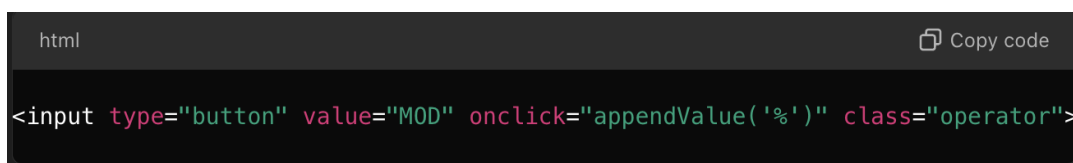
Features to Implement: While brainstorming, I reflected on features that would set my calculator apart. Drawing from my experiences with various calculators, I identified three key features:

- A modulus (MOD) button to appeal to computer science students like me.
- Visual feedback to confirm when buttons are clicked.
- A history feature to display previous calculations.

Feature Implementation:

1. Modulus Button:

I included a "MOD" button alongside the other arithmetic operators and ensured it worked as intended for modulus operations.

A screenshot of a code editor with a dark theme. The top bar shows 'html' on the left and a 'Copy code' button on the right. The code area contains a single line of HTML: `<input type="button" value="MOD" onclick="appendValue('%')" class="operator">`. The code is color-coded: 'type' is pink, 'button' is red, 'value' is green, 'MOD' is red, 'onclick' is pink, 'appendValue' is green, and '%' is red.

```
html Copy code
<input type="button" value="MOD" onclick="appendValue('%')" class="operator">
```

2. Visual Feedback:

To provide a responsive user experience, I implemented a highlight effect for buttons when clicked. Using a CSS class and a small JavaScript timeout, I ensured a smooth and

intuitive interaction.

```
css                                                                    Copy code

form input.clicked {
  background-color: #cab5bc; /* Highlight color */
}

javascript                                                             Copy code

button.classList.add('clicked');
setTimeout(() => {
  button.classList.remove('clicked');
}, 200); // Matches the CSS transition duration
```

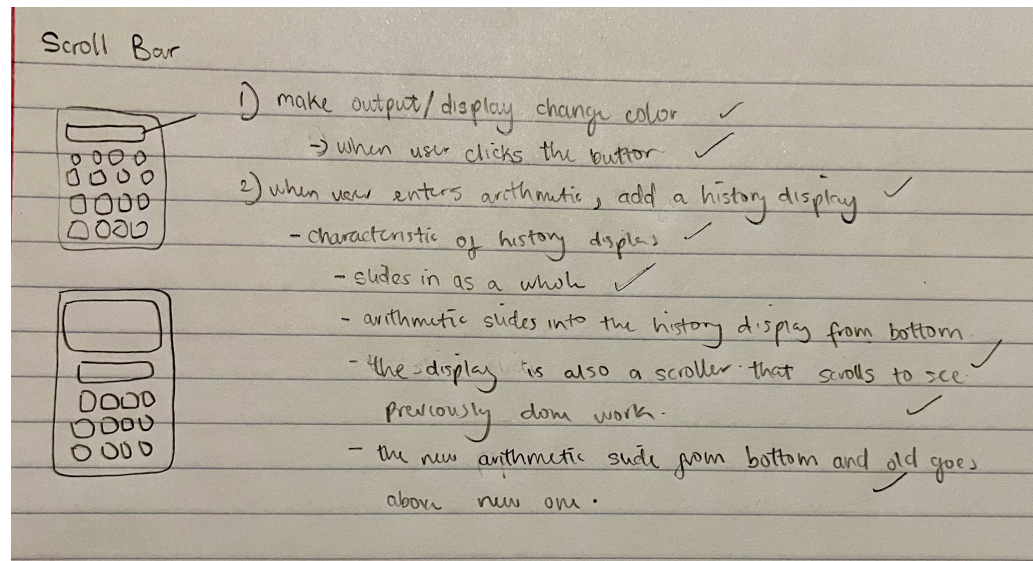
3. History Display:

I designed a scrollable history bar that updates with each calculation. The history is displayed in reverse order, making the most recent entry easily visible. A smooth slide-in animation enhances the user experience.

```
javascript                                                             Copy code

const historyEntry = document.createElement('div');
historyEntry.textContent = `${expression} = ${result}`;
historyEntry.classList.add('history-entry');
historyElement.appendChild(historyEntry);
historyElement.classList.add('visible');
historyElement.scrollTop = historyElement.scrollHeight;
```

- Low Fidelity Wireframe:



Visual Enhancements: To ensure the calculator was visually appealing, I animated the display and history bar. Initially hidden, the display appears once a button is pressed. Similarly, the history bar slides into view when calculations are made. While animating the history bar, I encountered a challenge where it would reanimate with each new entry. After debugging, I realized that I needed to restrict the animation trigger to initial visibility. The issue was resolved by managing class states effectively.

• Before Fix:

CSS

Copy code

```
.history-bar {
  animation: slideIn 0.5s ease-in-out;
}
```

• After Fix:

CSS

Copy code

```
.history-bar.visible {
  animation: none; /* Prevent reanimation */
}
```

Finally, I chose a color palette of Light pink and coral for the calculator's interface. These inviting colors evoke feelings of comfort, warmth, and hope, creating a naturally appealing design.



HEX **#F88379** RGB **248, 131, 121** HSL **5, 51%, 72%**



HEX **#FCE4EC** RGB **252, 228, 236** HSL **340, 10%, 94%**

Time Commitment: Approximately 35 hours

Through this project, I not only refined my technical skills but also explored the creativity and problem-solving aspects of design. It was a rewarding experience that further solidified my interest in web development.

Final Project link:

<file:///Users/daraajiboye/Documents/Personal%20Projects/New%20Calculator%20Project/index.html>